PHY 642

Special Topics in Quantum Mechanics

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NOTES TAKEN DURING 2025-26-I SEMESTER

These are notes taken from the lectures by **Prof. Suvankar Dutta**. I have tried to make them as comprehensive as possible with mathematics and diagrams wherever necessary.

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Course Content

I. Motivation, Review and Preparations

Context, applicability; decoherence is everywhere; single-particle quantum-mechanics, maths and statistical physics, classical stochastic process.

II. Basic Formalism and Interpretation of Decoherence

Quantum entanglement, density matrices, measurement problem, environmental monitoring, decoherence, dissipation and classical noise.

III. Master-Equation Formulation of Decoherence

Born-Markov masterequation, Lindblad Form, Non-Markovian dynamics, outlook on stochastic methods.

IV. Applications, Canonical Models for Decoherence

Two-level system, spontaneous decay, bath of harmonic oscillators, spin-Boson model, quantum Brownian motion, dissipation-fluctuation theorem.

V. Interpretations of Quantum Mechanics and the Quantum-to-Classical Transition

Interpretations, collapse theories, Bohmian mechanics.

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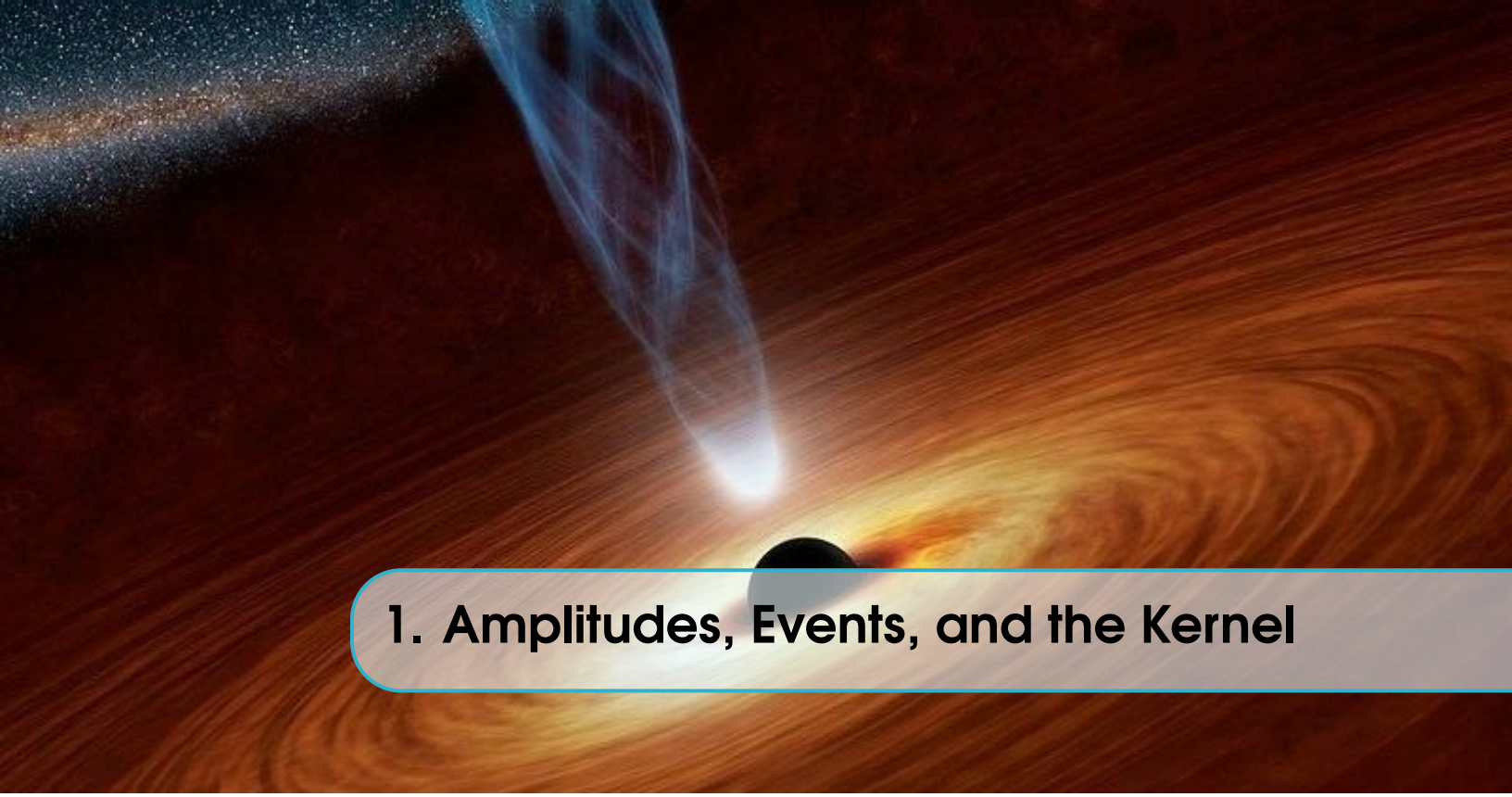

1. Amplitudes, Events, and the Kernel

This chapter summarizes the fundamental concepts of quantum probability, beginning with the idea of probability amplitudes and culminating in the path integral definition of the quantum-mechanical kernel.

1.1 The Foundation: Probability Amplitudes

The core departure from classical physics is in how probabilities are calculated. In experiments like the two-slit setup, the observed probability distribution, P, is not the sum of the probabilities for passing through each slit (P_1 and P_2).

Instead, quantum mechanics introduces a complex number called the **probability amplitude**, ϕ, for an event to occur. The probability is then found by taking the absolute square of this amplitude. The rules are as follows:

- The probability P of an event is the absolute square of its amplitude ϕ: $P = |\phi|^2$.
- If an event can occur in several alternative ways that cannot be distinguished by any experiment, the total amplitude ϕ is the **sum** of the amplitudes for each individual way ($\phi_1, \phi_2, \ldots$): $\phi = \phi_1 + \phi_2$.

1.2 Rules for Combining Amplitudes

Whether you add probabilities or amplitudes depends entirely on whether the alternative paths for an event can be distinguished. This leads to two types of alternatives.

1.2.1 Exclusive Alternatives

If an experiment is performed to determine which alternative is taken, the interference between them is destroyed. For example, if we use a light source to "watch" which slit an electron passes through, we find it unequivocally passes through one or the other. In this case, the alternatives are exclusive,

and the total probability is the sum of the individual probabilities:

$$P = P_1 + P_2$$

This happens because the act of observation fundamentally alters the outcome[cite: 314].

1.2.2 Interfering Alternatives

If no attempt is made to determine which alternative is taken, the paths are indistinguishable and the alternatives **interfere**. In this scenario, we must sum the probability amplitudes first, and then take the absolute square to find the final probability:

$$P = |\phi_1 + \phi_2|^2$$

This rule is fundamental to all of quantum mechanics and explains the wave-like interference patterns observed.

1.3 From Summing Alternatives to the Sum Over Paths

The concept of summing amplitudes can be extended from a few alternatives (like two slits) to an infinite number. We can imagine placing more screens between the source and the detector, each riddled with many holes. The total amplitude for a particle to arrive at the detector is the sum of the amplitudes for every possible path through the holes.

If we continue this process until the screens are completely gone, the sum over discrete paths through holes becomes an **integral over all possible paths** the particle could take through empty space. This conceptual leap is the foundation of the path integral formulation.

1.4 The Kernel as the Sum Over Paths

The ideas above are formalized in the concept of the **kernel**, or propagator, written as $K(b,a)$. This represents the total amplitude for a particle to get from a starting spacetime point $a = (x_a, t_a)$ to a final point $b = (x_b, t_b)$.

The kernel is defined as the sum of contributions from every possible path between the two points. This is expressed as the **path integral**:

$$K(b,a) = \int_a^b e^{(i/)S[x(t)]} \mathscr{D}x(t)$$

Where:

- $S[x(t)]$ is the **classical action** for a specific path $x(t)$. The action is the time integral of the Lagrangian.
- $e^{(i/)S[x(t)]}$ is the **probability amplitude** contributed by that single path. All paths contribute equal magnitude, but their phase is determined by the action.
- $\int_a^b \mathscr{D}x(t)$ symbolizes the **sum over all paths.**

Furthermore, amplitudes for events that occur in succession in time simply multiply. This leads to the crucial composition rule for kernels, where the total amplitude is found by integrating over all possible intermediate points x_c at an intermediate time t_c:

$$K(b,a) = \int_{-\infty}^{\infty} K(b,c)K(c,a)dx_c$$